GGR372 **Assignment 4**

Measuring Access and Neighbourhood Health Inequalities

Introduction:
The goal of this assignment is for you to learn how to perform basic measures of accessibility to measure health inequalities. In this assignment you will perform some accessibility analysis at the neighbourhood level. Using a spatial unit that is more widely appreciated by the public health and planning institutions can give rise to a more focused conversation about neighbourhood health inequalities.

Due: 35 marks and 15% of your final grade, due by March 27th at 5:00pm. Be prepared to show your work to your TA if requested.

Fast food density and low income neighbourhoods

Map 1: (10 Marks)
To begin you must locate the City of Toronto Neighbourhood Boundary file. To do so, go to Open Data Toronto (online) and find the file described as "Boundaries of the City of Toronto Neighbourhoods - historical 140". Download the WGS84 version and add it as a layer in ArcGIS, you will need this as your boundary map to display neighbourhood income (table provided). Be sure to use the historical 140 version of Toronto neighbourhoods.

In this hypothetical example you have been hired by Toronto Public Health to identify neighbourhoods that are of low income and have a high density of fast food restaurants. There are two steps to your analysis, first you need to calculate the density of fast food restaurants (per sq km) using the container approach. Second you need to create a thematic map for neighbourhood income in quartiles. Highlight the neighbourhoods that are in the highest quartile for fast food density as a layer with no fill. Then select and display the neighbourhoods that are in the highest quartile for fast food density and in the lowest quartile for neighbourhood income as a new layer with no fill in a different colour.

Questions:

1) How many neighbourhoods are classified as being the lowest income quartile and highest quartile for density of fast food restaurants? **1 mark**
2) Attach the table with the names of these neighbourhoods. **1 mark**

Your final map should show the following elements

- Neighbourhoods with the highest density of fast food restaurants (based on quartile classification)
- Neighbourhoods with the highest density of fast food restaurants and lowest income neighbourhoods (both based on quartile classification)
- A thematic map of median household income by neighbourhood
- Any other reference data that was given to you that may be appropriate

Sources:
City of Toronto, 2016, DMTI 2016, Census of Canada 2016.

Notes:
These datasets have not been verified. The analysis you are completing is meant to be a useful exercise, but the results may not be an accurate representation of fast food access in Toronto.

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Map 2: (10 Marks)

For the second part of your study, you have been asked to look at access to fast food restaurants around secondary schools. Using the origin destination cost matrix in Network Analyst calculate the distance from each of the secondary schools to the nearest three fast food restaurants. Remember the Unique ID field in the school file is extremely important, as you will have to join the results back to the schools point file by this field. I suggest using a search tolerance of 500 metres. Be sure to include Unique ID as the name field when loading your origins. Remember to only calculate the distance to the nearest 3 destinations. After calculating the distance to the nearest 3 fast food outlets, you will need to use the summarize tool to determine the average distance to 3 fast food restaurants. Remember, the Unique ID field is essential here again. As we discussed in class, you may need to manipulate the Name field after the analysis to get your Unique ID field. Please create a map that shows the schools with easy walking access to fast food restaurants. Easy access should be classified as an average walking distance of 500 metres or less to three fast food restaurants. Also show this on a thematic map for neighbourhood income in quartiles. Please display the neighbourhoods within the lowest income quartile that have a school with easy fast food access with no fill.

Your final map should show the following elements

- Schools with easy access (average distance of 500 metres or less to 3 fast food restaurants)
- Neighbourhoods with a school with easy fast food access and lowest income neighbourhoods (based on quartile classification)
- A thematic map of median household income by neighbourhood
- Any other reference data that was given to you that may be appropriate

Questions:

1) How many lines were created in the OD cost matrix? **1 mark**
2) How many schools are within an average distance of 500 metres to three fast food restaurants? **1 mark**
3) How many of these schools are also in a neighbourhood classified as the lowest income (quartile classification)? **1 mark**

Report: (10 marks)

Write a brief report to explain the methods you used and highlight your findings from the two studies. Your report must reference your maps, state the important steps taken and include some writing explaining how fast food access relates or does not relate to neighbourhood income. Discuss fast food access within the context of socio-spatial inequities. (Maximum 200 words)

Methods: 5 marks
Discussion: 3 marks
Overall report: 2 marks

Data provided:

Fast food restaurants.shp
Secondary schools.shp
Network dataset (walking network)
Neighbourhood income (table)
Other reference data (i.e., Lake Ontario, surrounding areas)